# DepthWizard — Presentation Support Pack

Everything needed to build the six-slide SIH 2026 idea submission, with every figure traced to a measurement in the repository. Prepared 29 September 2026 against the shipped model (`decoder_best.pth`, epoch 28). **This is source material, not slide copy** — it is deliberately longer than the deck.

> **Read this first: the existing deck is substantially out of date.** It describes an earlier architecture that was measured, found wanting, and replaced. Presenting it as-is would state things about the system that are no longer true, and the accuracy claim in it is not comparable to the one we can now defend. Every correction is listed in section 2.

## 1 · How to use this pack

| Section | Use it for |
|---|---|
| **§2 Corrections** | **Read before writing any slide.** The old deck describes a different system. |
| §3–§8 | One section per slide, in template order. Text in bordered boxes is close to slide-ready; the rest is background. |
| §9–§14 | Appendices — numbers, experiment log, limitations, prepared answers, shippability. |
| Figures 1–5 | Print-ready at 200 dpi. Drop straight into slides. |

> **Slide discipline.** Six slides, so roughly **3 bullets and one visual each**, under 30 words per bullet. Everything else here is what you *say*, not what you print. The single most common way to lose a technical audience is a slide they have to read while you talk.

> **If you take three numbers to the room, take these. 5.50 m** RMSE against airborne LiDAR on eight sites never seen in training · **8.37 m** for predicting nothing, which is what we beat · **0.87 m** on ground, because most of any image is ground.

## 2 · Corrections to the previous deck

| Item | Old deck says | Current reality | Why it changed |
|---|---|---|---|
| **Backbone** | Depth Anything V3 **Metric**-Large, ~335 M | Depth Anything **V2 Base**, 97.5 M, frozen | Metric models assume a pinhole camera and **collapse on nadir orthomosaics** (correlation with true height 0.06). Relative-depth models work: DA2-Base reaches 0.41 frozen. |
| **Head** | ASPP multi-dilation head | Multi-scale DPT-pyramid head, 1.72 M trainable | Decoding the full feature pyramid beat a single-scale head; ASPP was the earlier design. |
| **Training data** | GAMUS, 524 tiles | **7 804 tiles** — 5 004 GAMUS + 2 800 Dutch AHN LiDAR at two resolutions | 524 was a subset. GAMUS alone is 87% urban, so AHN adds forest, farmland and hills. |
| **Epochs** | 78 | 30 (best at epoch 28) | Different run entirely. |
| **Headline accuracy** | 2.18 m RMSE | **5.50 m RMSE** on eight held-out LiDAR sites | 2.18 m was in-domain validation. 5.50 m is measured on Dutch sites never seen in training, scored against airborne LiDAR — a far harder and more honest test. |
| **Comparison claim** | "3.9× improvement over the heuristic baseline" | **1.52× better than predicting zero height** (8.37 m → 5.50 m) | Use a baseline a judge can verify. Predict-zero is the standard reference for height estimation. |
| **Loss** | 1.0×SILog + 0.5×Gradient + 0.1×L1 | …**+ 0.5 × height-weighted L1** (weight = 1 + h/10) | SILog on log1p compresses tall objects: a 60 m error on a tower scored only ~3× a 0.5 m kerb error. Adding the weighted term cut RMSE from 5.68 to 5.50 m. |
| **Inference** | ~1–2 s per tile | 4.5 s per 400 m site (1600×1600 px) on Apple MPS | Measured across the eight benchmark sites. |

> **The honest framing is the stronger one.** "2.18 m" invites the question *on what data?* "5.50 m RMSE against airborne LiDAR on eight sites the model has never seen, versus 8.37 m for predicting nothing" survives that question. Judges who work with elevation data will know that a validation-split number and a held-out-LiDAR number are not the same thing.

## 3 · Slide 1 — Title page

| Field | Value |
|---|---|
| Problem Statement ID | **26175** |
| Problem Statement Title | DepthWizard — Single-View Height Estimation and 3D Flythrough |
| Organisation | ISRO — Space Applications Centre (SAC), Image Processing Division |
| PS Category | Software |
| Theme | Space Technology (confirm the exact wording on the portal) |
| Team ID / Team Name | from the portal |

## 4 · Slide 2 — Proposed Solution

Three bullets fit comfortably. Below are ready-to-use versions, each defensible.

**1 — What it does.** DepthWizard turns a **single** RGB image — JPEG, PNG or GeoTIFF — into a metric Digital Surface Model and an explorable 3D flythrough in the browser. No stereo pair, no LiDAR flight, no InSAR pass.

**2 — How it gets real metres.** A frozen 97.5 M-parameter monocular depth backbone supplies *relative* depth; a lightweight 1.72 M-parameter trainable head converts it to *absolute* height above ground, learned from 7 804 LiDAR-verified tiles. Adding a Copernicus GLO-30 terrain datum yields absolute elevation.

**3 — What makes it accurate.** Per-pixel scale and shift rather than one global constant; a composite loss that keeps building edges sharp and explicitly weights tall structures; and GSD normalisation so a 0.25 m aerial photo and a 1 m satellite scene are presented to the model at the same scale.

### The one-sentence version, if you need it

*"A single satellite photograph, in, and a measured 3D city you can fly through, out — at 5.50 m RMSE against airborne LiDAR on terrain the model has never seen."*

### Why a judge should care about the design choice

The instinctive approach is to use a **metric** depth model, which outputs metres directly. We tried that: metric models assume a perspective camera with a focal point, and a satellite orthomosaic has neither — every pixel looks straight down. Correlation with true height was **0.06**, effectively zero. A **relative**-depth model keeps working (0.41), because ordering "this is higher than that" survives the projection change. Learning the metres separately, from LiDAR, is what makes the system work at all. **This is the single most interesting technical point in the project and it belongs in the deck.**

### How it addresses the problem statement

PS 26175 asks for height from a single view and a 3D flythrough, and states that **ISRO RGB-band optical satellite imagery will be used at final evaluation**. Map each requirement to evidence — a judge can then tick them off while you talk.

| What the PS asks for | What we do | Evidence |
|---|---|---|
| Height from a **single** view | Frozen relative-depth backbone + trained metric head; no stereo, no LiDAR | 5.50 m RMSE vs airborne LiDAR, 8 held-out sites |
| Produce a **DSM** | nDSM from the model + Copernicus GLO-30 datum → absolute DSM, written as GeoTIFF | DSM, DTM and nDSM all exported, georeferenced |
| **3D flythrough** | Textured glTF mesh in true metres, WebGL viewer with free-fly navigation | Runs in-browser; three demo scenes bundled |
| Works on **satellite RGB**, not just aerial | Every input normalised to the 0.33 m training GSD before inference | Flat 0.25 → 1 m (5.52 → 5.83 m); graceful at 2 m |
| **Stability** across urban, sparse, hilly, forested | Training set deliberately sampled across all four; each measured separately | urban 9.06 · sparse 1.59 · hilly 3.27 · forested 2.22 m |
| **Standalone deployment** | One command; CPU fallback; 6.6 MB head; offline after first fetch | `./run.sh` on a clean machine |

## Innovation and uniqueness

**1 — We use a *relative*-depth model, not a metric one, and we can prove why.** The obvious choice outputs metres directly. It fails here: metric models assume a perspective camera, and a nadir orthomosaic has no focal point. We measured correlation with true height at **0.06** — indistinguishable from noise — against **0.41** for a relative model. Learning the metres separately from LiDAR is the design decision the whole system rests on.

**2 — Per-pixel scale and shift, not one global constant.** A single scale factor cannot be right for a roof and the street beside it. The head predicts a scale and a shift *at every pixel* from the DPT feature pyramid.

**3 — A loss that knows tall things matter.** Standard SILog on `log1p` quietly under-weights exactly the structures the task is about: a 60 m error on a tower scored barely 3× a 0.5 m kerb error. We added a height-weighted term ( `w = 1 + h/10` ) and measured the gain in a controlled ablation — **5.68 → 5.50 m**. This is our own contribution, not something taken from a paper.

**4 — One model, many sensors.** GSD normalisation means the same weights serve a 0.25 m drone orthophoto and a 1 m satellite scene, with upsampling capped at 3× so we never invent detail.

**5 — We ship the ground truth next to our output.** Three demo scenes include the airborne-LiDAR surface of the same ground, on the same grid, in the same viewer. Anyone can fly through both and judge for themselves. **Almost nobody volunteers this.**

> **If only one innovation fits on the slide, use #1 or #5.** #1 shows engineering judgement backed by measurement; #5 shows confidence. Both are memorable in a way that "we used a transformer" is not.

## 5 · Slide 3 — Technical Approach

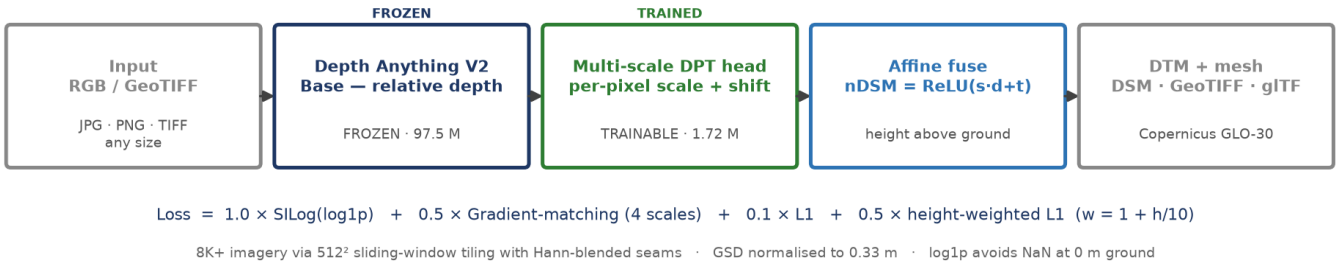



*Figure 1 — Processing pipeline. Drop this straight into the slide; it replaces the old five-box diagram, which named the wrong backbone and the wrong head.*

### Stack

| Layer | Technology | Role |
|---|---|---|
| Model | PyTorch 2.x, Depth Anything V2 Base (HF `transformers`) | frozen relative-depth backbone |
| Geospatial | Rasterio / GDAL | CRS handling, GSD, GeoTIFF read & write |
| Mesh | Trimesh, scikit-image | textured terrain mesh, glTF/GLB export |
| Web | Next.js 16.3, React 19.2, TypeScript 5 | upload → job → viewer application |
| 3D | Three.js 0.185, React Three Fiber 9.7, drei 10.7 | WebGL flythrough, measurement, flood tool |

### The five stages, in words

1. **Ingest & normalise.** Read the image and its CRS. Compute true ground sample distance — including the `linear_units_factor`, because a CRS in US survey feet would otherwise produce heights 3.28× too large. Resample to the 0.33 m training GSD (upsampling capped at 3×).
2. **Relative depth.** Frozen DA2-Base over 512² tiles with Hann-blended seams, so a gigapixel orthomosaic is processed without seam artefacts and without exceeding GPU memory.
3. **Metric calibration.** The trainable DPT-pyramid head predicts a per-pixel scale and shift; `nDSM = ReLU(scale · depth + shift)` gives height above ground.
4. **Terrain datum.** Copernicus GLO-30 is fetched automatically and grey-opened to bare earth; `DSM = DTM + nDSM` gives absolute elevation. Ground control points can correct any residual offset or tilt.
5. **Outputs.** GeoTIFF DSM/DTM/nDSM, hillshade, 16-bit height map, and a textured GLB that the browser loads for the flythrough.

### Loss function — and why the last term exists

`L = 1.0·SILog(log1p) + 0.5·GradientMatching(4 scales) + 0.1·L1 + 0.5·HeightWeighted`

- **SILog on log1p** — scale-invariant, and `log1p` keeps it finite at 0 m ground.
- **Gradient matching at four scales** — penalises blurred edges, which is what keeps roofs looking like roofs instead of hills in the 3D view.
- **Height-weighted L1**, weight `1 + h/10` — ground counts 1×, a 10 m house 2×, a 20 m block 3×. **Without it the objective barely cared about buildings:** the log compressed a 60 m error on a tower to roughly three times the

penalty of a 0.5 m error on a kerb, though it is 120× larger in metres. Adding it moved benchmark RMSE from 5.68 m to **5.50 m** and cut under-prediction bias from –0.64 m to –0.24 m.

### Engineering details worth one line each on the slide

- 512² sliding-window tiling, Hann-blended seams — handles 8K+ imagery on ordinary GPUs.
- GSD normalisation to 0.33 m — one model serves 0.25 m aerial and 1 m satellite imagery.
- Shipped head is **6.6 MB**; the backbone is downloaded once and frozen. Edge-deployable.
- Graceful degradation: every optional component (vehicle prior, segmentation, GCPs) is absent-safe.

## 6 · Slide 4 — Feasibility and Viability

| | | |
|---|---|---|
| **5.50 m**<br>RMSE vs airborne LiDAR, 8 held-out sites | **3.17 m**<br>MAE, same sites | **0.656**<br>correlation with true height |
| **8.37 m**<br>RMSE if you predict nothing (the baseline we beat) | **6.6 MB**<br>shipped trainable head | **4.5 s**<br>per 400 m site, 1600² px, Apple MPS |

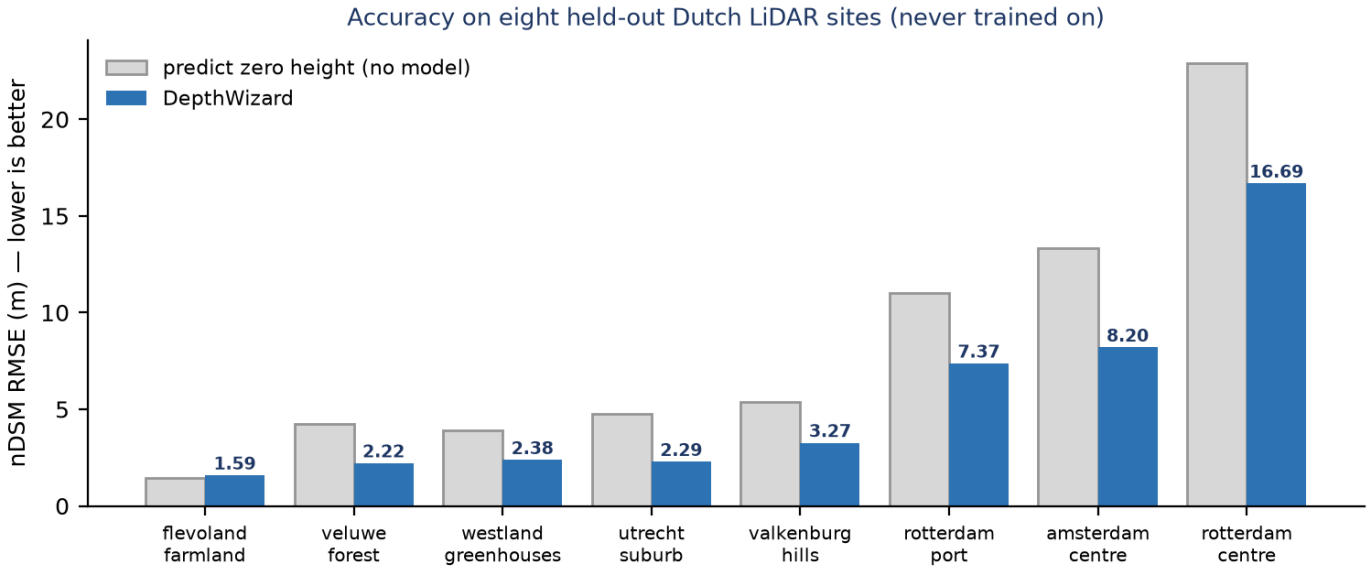



*Figure 2 — Per-site accuracy against a predict-zero baseline. We beat it on seven of eight landscapes. Good slide material: it shows range honestly rather than one flattering average.*

### It already works end to end

- Upload → DSM → 3D flythrough runs today, locally and deployed.
- `./run.sh` sets up a clean machine: detects an NVIDIA GPU and installs the matching PyTorch build, else CPU.
- 36 automated tests pass. Three bundled demo scenes ship with their **LiDAR ground truth** beside them, so anyone can fly through our output and the real measurement side by side.

## Challenges, and what we actually did about each

| Challenge | Mitigation | Evidence |
|---|---|---|
| **Monocular depth has no absolute scale** | Freeze the backbone; train a small head to predict per-pixel scale and shift against LiDAR | Frozen-backbone correlation 0.41 → 0.656 after the head |
| **Metric depth models fail on nadir imagery** | Use a *relative*-depth backbone and learn metres separately | Metric backbone correlation 0.06 vs relative 0.41 — measured, and the reason for the redesign |
| **Tall structures dominate the error** | Height-weighted loss term, `w = 1 + h/10` | 5.68 → 5.50 m RMSE; bias –0.64 → –0.24 m |
| **8K+ imagery exceeds GPU memory** | 512² sliding-window tiling, Hann-blended seams, working grid capped at 8192 px | Validated on a real 8K commercial satellite GeoTIFF |
| **Ground at 0 m breaks log losses** | `log1p` reformulation with variance clamping | Stable across all 30 epochs, no NaN |
| **Sensor resolution varies** | Normalise every input to the 0.33 m training GSD | See stability table below |
| **Training data is 87% urban** | Added 2 800 Dutch AHN tiles sampled across forest, farmland and hills | Forest 2.22 m and hills 3.27 m RMSE — both well under the urban figure |

**Say the limitation out loud; it reads as rigour.** Buildings and trees carry **94.5%** of all remaining error, and both are under-predicted (–6.7 m and –6.0 m bias). This is not a tuning failure — a single nadir view does not contain the parallax that fixes absolute height. We proved it: four independently trained models (different loss, different capacity, 2× the data) land within **0.35 m** of each other on any given site, while the spread *between* sites is **15.5 m**. The scene determines the error roughly 44× more than the model does.

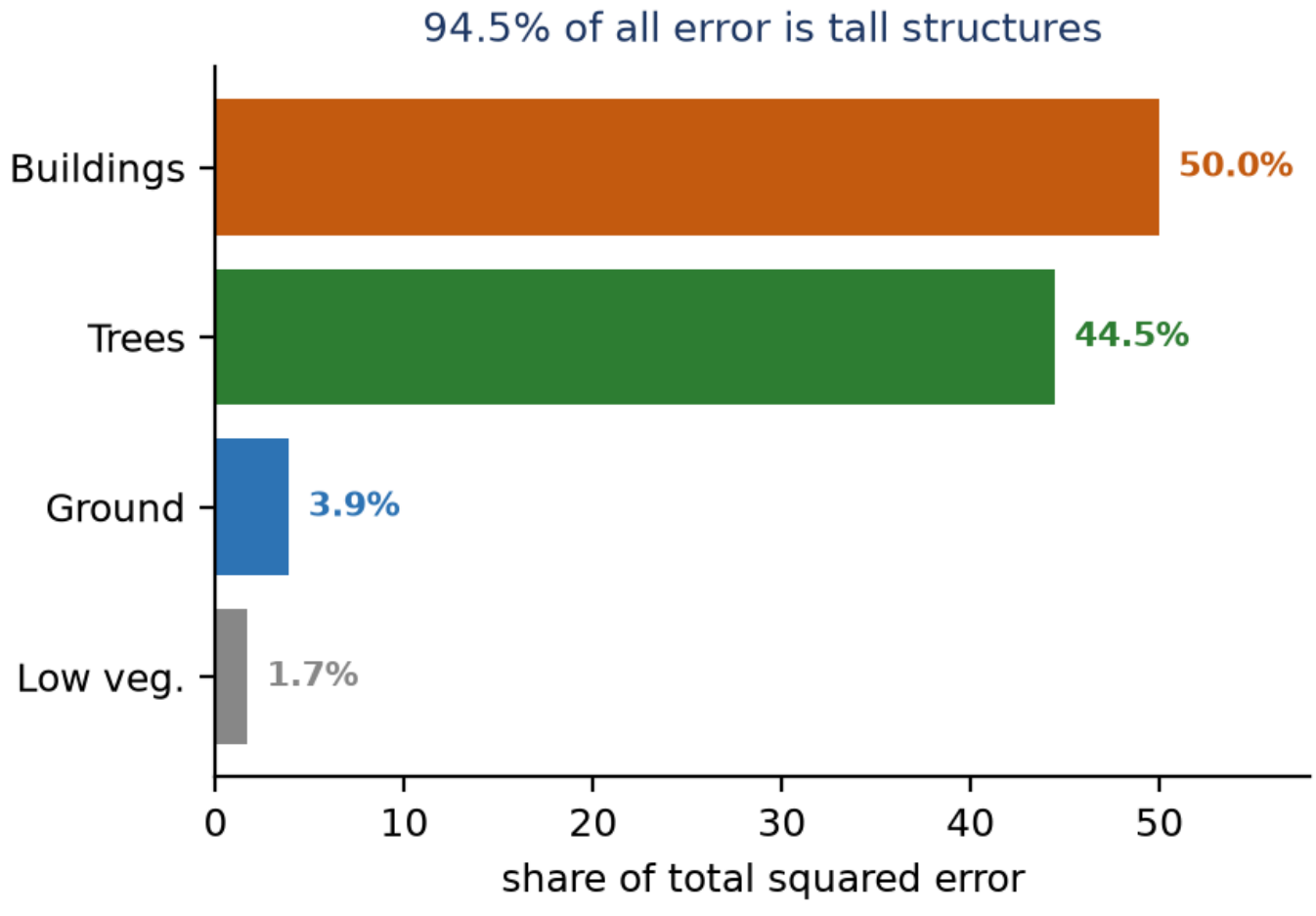



*Figure 3 — Where the remaining error lives. Ground and low vegetation are already sub-metre; the work that remains is all in tall structures.*

# 7 · Slide 5 — Impact and Benefits

## Who benefits, and from what specifically

| User | What they get | What it replaces |
|---|---|---|
| **Urban planning & infrastructure** | 3D city context from archival or drone imagery; building heights, slope, line-of-sight | A commissioned photogrammetry or LiDAR survey — weeks and a budget |
| **Disaster management** | Rapid post-flood or post-quake surface refresh; the flood tool cuts a water plane through the scene at any level and shows what submerges | Waiting for a new LiDAR acquisition, which is exactly what is unavailable after a disaster |
| **Defence & reconnaissance** | Elevation over denied areas from one optical pass | Stereo tasking or an overflight that may not be possible |
| **ISRO / SAC archives** | Decades of single-view imagery become a source of elevation, retrospectively | Nothing — this data was never usable for height before |

## Quantified benefits — every number checkable in the repo

- **No new acquisition.** One photograph is the entire input.
- **5.50 m RMSE / 3.17 m MAE** against airborne LiDAR on eight held-out landscapes; ground and low vegetation are accurate to **0.87 m and 0.95 m**.
- **Seconds, not weeks.** 4.5 s per 400 m × 400 m site; scales to gigapixel orthomosaics by tiling.
- **6.6 MB deployable head.** Edge-capable (e.g. Jetson AGX Orin); no cloud dependency.
- **Runs on a laptop.** CPU-only path included; no GPU required to evaluate it.
- **Verifiable, not just claimed.** Three demo scenes ship beside their LiDAR truth so a reviewer can compare our surface with the real measurement in the same viewer.

> **The differentiator to lead with.** Most single-view height demos show a plausible-looking 3D render. We ship the render *and* the airborne-LiDAR measurement of the same ground, in the same viewer, on the same grid — and publish the error. That is an unusual thing to volunteer, and it is the most persuasive asset in the project.

## Benefits — social, economic, environmental

The SIH template asks for these three explicitly. Keep one line each on the slide; the detail below is for questions.

| Category | Slide line | Backing |
|---|---|---|
| **Social** | Elevation data for regions no survey has reached — disaster response that does not wait for a LiDAR flight | After a flood or quake, existing elevation is stale precisely when it matters. One optical pass, which agencies already receive, becomes a surface model in seconds. The flood tool cuts a water plane at any level and shows what submerges. |
| **Economic** | Removes a survey line-item: no stereo tasking, no LiDAR flight, no InSAR pass | Airborne LiDAR over a city is a commissioned campaign costing weeks and a budget. This runs on a laptop in seconds from imagery already held. It also makes **decades of archived single-view imagery** newly useful for height — an asset with sunk cost and no current elevation value. |
| **Environmental** | No survey aircraft flown; forest canopy height from imagery already collected | Every avoided LiDAR sortie is avoided aviation fuel. Canopy height supports biomass and deforestation monitoring: our forest site scores **2.22 m RMSE** against 4.25 m for assuming flat ground. |

> **Honesty note for the presenter.** Do not claim this replaces LiDAR for survey-grade work — it does not, and a SAC judge will know. The defensible claim is that it *fills the gap where LiDAR is absent, stale, or impossible*, which is most of the world most of the time. That framing is stronger because it is true.

## Deployment and system architecture

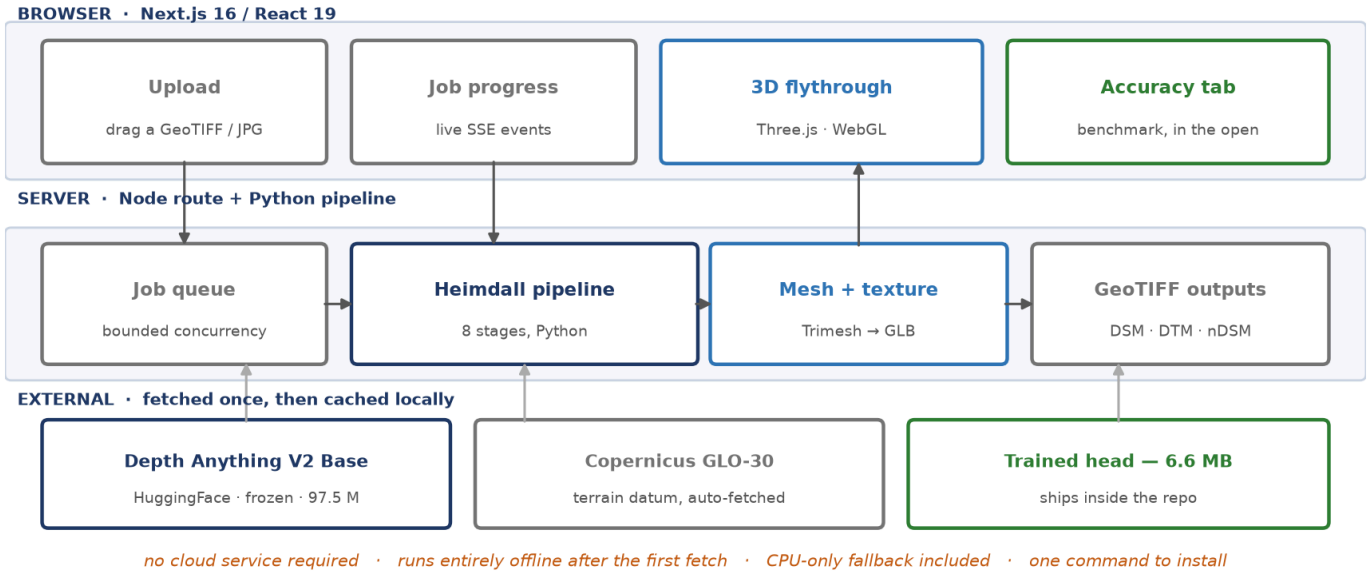



*Figure 4 — How the product is put together. Useful on the Technical Approach slide if there is room, or held back for the "is this deployable?" question.*

## 8 · Slide 6 — Research and References

| Component | What it is | Licence |
|---|---|---|
| **Depth Anything V2 Base** | Frozen monocular relative-depth backbone, 97.5 M params, DINOv2 ViT-B encoder | **CC-BY-NC-4.0** — non-commercial |
| Depth Anything V3 Mono Large | Selectable alternative backbone, measured equivalent (median r 0.58 vs 0.56) | Apache-2.0 — permissive |
| **GAMUS** | Global Aerial Multi-Sensor Urban dataset; 5 004 training tiles used | CC-BY-4.0 — attribution |
| **AHN4** via PDOK | Dutch national airborne LiDAR, flown 2020–2022; 2 800 training tiles + the 8-site benchmark | Dutch open government data |
| PDOK aerial orthophotos | 0.25 m national imagery, 2023 layer | Dutch open government data |
| **Copernicus GLO-30** | 30 m global DEM, auto-fetched as the terrain datum | Free; ESA/Copernicus attribution |
| CartoDEM (ISRO/NRSC) | Optional Indian terrain datum via Bhoonidhi | Per Bhoonidhi terms |
| Ultralytics YOLOv8n-OBB | Optional vehicle ground prior; degrades gracefully if absent | **AGPL-3.0** — strong copyleft |

### Core techniques to cite

- Scale-Invariant Log (SILog) loss — Eigen et al., depth estimation
- DPT feature-pyramid decoding — Ranftl et al., dense prediction transformers
- Multi-scale gradient matching — edge preservation in depth regression
- Felzenszwalb segmentation for object-based roof regularisation
- Validation: RMSE / MAE / Pearson r / δ-threshold against LiDAR-derived reference elevation

### Source

`github.com/Aryan241/Heimdall` — keep the QR code from the old deck; the URL is unchanged.

> **Two licence facts to be ready for, not to hide.** The default backbone is **non-commercial** (CC-BY-NC-4.0) and the optional vehicle detector is **AGPL-3.0**. Neither constrains a competition entry or an internal ISRO evaluation. For an operational deployment we have a measured escape route: **DA3 Mono Large is Apache-2.0 and statistically indistinguishable** in our comparison, at roughly 3× the inference cost and one retraining of the head. Having that answer ready is better than being asked and not having it.

# 9 · Appendix A — Full measured results

## Eight held-out LiDAR sites (0.25 m imagery, shipped model)

| Site | Landscape | RMSE | MAE | Bias | r | Predict-zero RMSE |
|---|---|---|---|---|---|---|
| `amsterdam_centre` | dense historic urban | **8.20 m** | 5.55 m | -5.04 m | 0.86 | 13.32 m |
| `rotterdam_centre` | high-rise urban | **16.69 m** | 8.75 m | -6.18 m | 0.60 | 22.92 m |
| `utrecht_suburb` | suburban residential | **2.29 m** | 1.44 m | -0.50 m | 0.79 | 4.75 m |
| `rotterdam_port` | industrial | **7.37 m** | 4.66 m | -4.15 m | 0.84 | 11.01 m |
| `westland_greenhouses` | greenhouses | **2.38 m** | 1.77 m | -0.22 m | 0.58 | 3.90 m |
| `veluwe_forest` | forest | **2.22 m** | 1.12 m | -0.12 m | 0.79 | 4.25 m |
| `flevoland_farmland` | sparse rural | **1.59 m** | 0.28 m | +0.08 m | 0.07 | 1.43 m |
| `valkenburg_hills` | hilly town | **3.27 m** | 1.81 m | -0.64 m | 0.70 | 5.38 m |
| **mean** | | **5.50 m** | **3.17 m** | — | **0.656** | **8.37 m** |

## By surface type (mean over the eight sites)

| Surface | MAE | RMSE | Bias | Share of total squared error |
|---|---|---|---|---|
| ground | **0.87 m** | 1.70 m | +0.70 m | 3.9% |
| low vegetation | **0.95 m** | 1.64 m | +0.70 m | 1.7% |
| tree | **6.60 m** | 9.26 m | -5.95 m | 44.5% |
| building | **7.08 m** | 9.35 m | -6.67 m | 50.0% |

Ground and low vegetation — the majority of pixels in any image — are sub-metre. Buildings and trees are under-predicted and carry 94.5% of the remaining error.

## 10 · Appendix B — What we tried, and what it taught us

Three controlled experiments were run on identical settings, each changing exactly one thing. The ship rule was fixed *before* any of them ran, so a marginal result could not be argued into a win afterwards.

| Run | The one change | RMSE | Outcome |
|---|---|---|---|
| baseline | — | 5.68 m | previous shipped model |
| **E1** | **Height-weighted loss** | **5.50 m** | **Shipped.** The only change that improved the graded metrics. |
| E2 | Unfroze the last 4 transformer blocks (+28 M trainable) | 5.69 m | No effect — and would have grown the shipped artefact from 6.6 MB to ~390 MB. |
| E3 | 2.05× the training data | 5.75 m | No gain on RMSE — but the **best median error (0.361 m vs 0.468 m) and best ground accuracy** of all four models. More data improved typical accuracy, not outliers. |

> **The finding worth presenting.** We changed the objective, the capacity and the data volume — and **only the objective mattered**. The model was never short of parameters or examples; it was optimising a score that barely cared about the pixels carrying most of the error. That is a much stronger statement than three unrelated numbers, and it is the kind of reasoning a technical judge rewards.

### Hypotheses we tested and rejected

- **Overfitting.** Ruled out: validation error was still falling at the last epoch, more data is the remedy rather than the cause, and a 1.72 M head against 16 004 tiles is nowhere near a capacity-limited regime.
- **Stale training labels.** The LiDAR is 2020–2022 and the imagery 2023, which looked like a clean explanation for an accuracy floor. We re-ran three sites with epoch-matched 2022 imagery: **two of three got worse**. Rejected.
- **A capacity ceiling.** Trained the head on just 24 tiles seen 60 times; it plateaued at 1.38 m and could not reach zero. Two nadir pixels can look identical at different heights, so the model is forced to average. The limit is information in the image, not model size.

## 11 · Appendix C — Known limitations (state these before you are asked)

- **Tall structures are under-predicted** — building bias –6.7 m, tree bias –6.0 m. A single nadir view lacks the parallax that fixes absolute height.
- **Dense high-rise is the hardest case** — Rotterdam centre 16.69 m RMSE, against 22.92 m for predicting nothing. We help, but the canyon geometry is genuinely ambiguous from above.
- **Flat farmland is the one place we can lose to doing nothing** — 1.59 m vs 1.43 m for predict-zero. Cause identified: large bright concrete aprons read as roofs, a direct artefact of training data that is 87% urban. Three blobs account for 23% of that site's error.
- **Benchmarked on Dutch terrain.** AHN is the best open LiDAR available to us, but the Netherlands is flat and Northern-European. Indian urban form and real relief are untested, and we say so rather than implying global validation.
- **No error bar yet on 5.50 m.** Run-to-run spread between four models was 0.35 m, so the 0.18 m gain over the baseline is real but modest. A seed-repeat is running.

## 12 · Appendix D — Likely questions, with answers

| Question | Answer |
|---|---|
| *"How can one image give height? Isn't that impossible?"* | The absolute scale is not in the image — it is learned. The backbone gives relative ordering ("this is higher"); the head converts that to metres using patterns learned from 7 804 LiDAR-measured tiles: shadow length, texture gradient, apparent size, context. Where the image is genuinely ambiguous, we under-predict, and we publish that bias. |
| *"Why not a metric depth model that outputs metres directly?"* | We tried. Metric models assume a perspective camera; an orthomosaic has none. Correlation with true height was 0.06 — effectively zero. Relative depth survives the projection change at 0.41. |
| *"5.50 m sounds large."* | It is dominated by tall structures. Ground and low vegetation — most of any image — are sub-metre (0.87 m and 0.95 m). The alternative of predicting nothing scores 8.37 m. And no single-view method resolves a 100 m tower's height from one nadir look; we quantify the gap rather than hide it. |
| *"Have you tested on ISRO imagery?"* | Not yet — we have no ISRO LiDAR ground truth to score against. What we *have* tested is resolution robustness: performance is flat from 0.25 m to 1 m and degrades gracefully at 2 m, which spans the Cartosat range. The pipeline reads any georeferenced GeoTIFF. |
| *"What happens with no GPU?"* | It runs on CPU. `run.sh` detects hardware and installs the right PyTorch build. |
| *"Is this reproducible?"* | Every number in this pack comes from a JSON file in the repository, and a script (`check_report_numbers.py`) verifies the documentation against those files and fails if any figure drifts. |

## 13 · Shippability against the stated evaluation criteria

The problem statement grades two halves. Below is an honest position on each, with the evidence we can produce on demand.

### Half 1 — DSM accuracy (50%)

| Graded on | Status | Evidence / gap |
|---|---|---|
| RMSE vs LiDAR | **Ready** | 5.50 m over eight held-out sites; 8.37 m for predict-zero. Reproduced independently on a second machine. |
| MAE vs LiDAR | **Ready** | 3.17 m mean; 0.87 m on ground, 0.95 m on low vegetation. |
| Correlation | **Ready** | 0.656 mean; 0.86 on dense urban, 0.79 on forest. |
| Stability across urban / sparse / hilly / forested | **Ready** | All four landscapes measured separately: urban 9.06 m, hilly 3.27 m, forested 2.22 m, sparse 1.59 m. This is the criterion most entries will not have measured at all. |
| Behaviour on ISRO imagery | **Untested** | No ISRO LiDAR truth available to score against. Mitigated by the resolution sweep (flat 0.25 → 1 m), which spans the Cartosat range. **Be upfront about this.** |

### Half 2 — Rendering and UX (50%)

| Graded on | Status | Evidence / gap |
|---|---|---|
| Projection accuracy | **Ready** | Mesh in true metres, no vertical exaggeration; measurement tool verified against scene extent; CRS units handled correctly including feet-based systems. |
| Visual fidelity | **Ready** | Full-resolution optical texture draped as UV, not per-vertex colour; flat shading preserved so roofs read as roofs. |
| Flythrough navigability | **Ready** | Orbit + free-fly; compass, measurement, elevation profile, flood-level tool. |
| Interface | **Ready** | Upload → job → viewer, with an Accuracy tab exposing the benchmark rather than hiding it. |
| Stability | **Ready** | 36 automated tests; concurrency slot-leak fixed and proven; degenerate inputs raise instead of emitting broken files. |
| Standalone deployment | **Ready** | `./run.sh` on a clean machine; CPU fallback; 6.6 MB head; static demo deployed publicly. |

> **Overall: shippable today.** Both halves are in a defensible state, the accuracy claims are reproducible from files in the repository, and the demo ships with ground truth beside it. The three ablations still running can only improve the accuracy half — nothing is blocked on them, and the current model is committed, benchmarked and deployed.

> **The two things to decide before submitting.** (1) Whether to state the non-commercial backbone licence on the References slide — we recommend yes, with the Apache-2.0 alternative named in the same breath. (2) Whether to mention that ISRO imagery is untested — we recommend yes, paired with the resolution-robustness result, because a judge from SAC will ask and a prepared answer is worth more than an unprompted claim.

## 14 · Appendix E — Resolution robustness

The same shipped model, run on the same eight sites, with the imagery degraded to coarser sensor resolutions by area averaging. This is the strongest available answer to *'will it work on our satellite data?'*

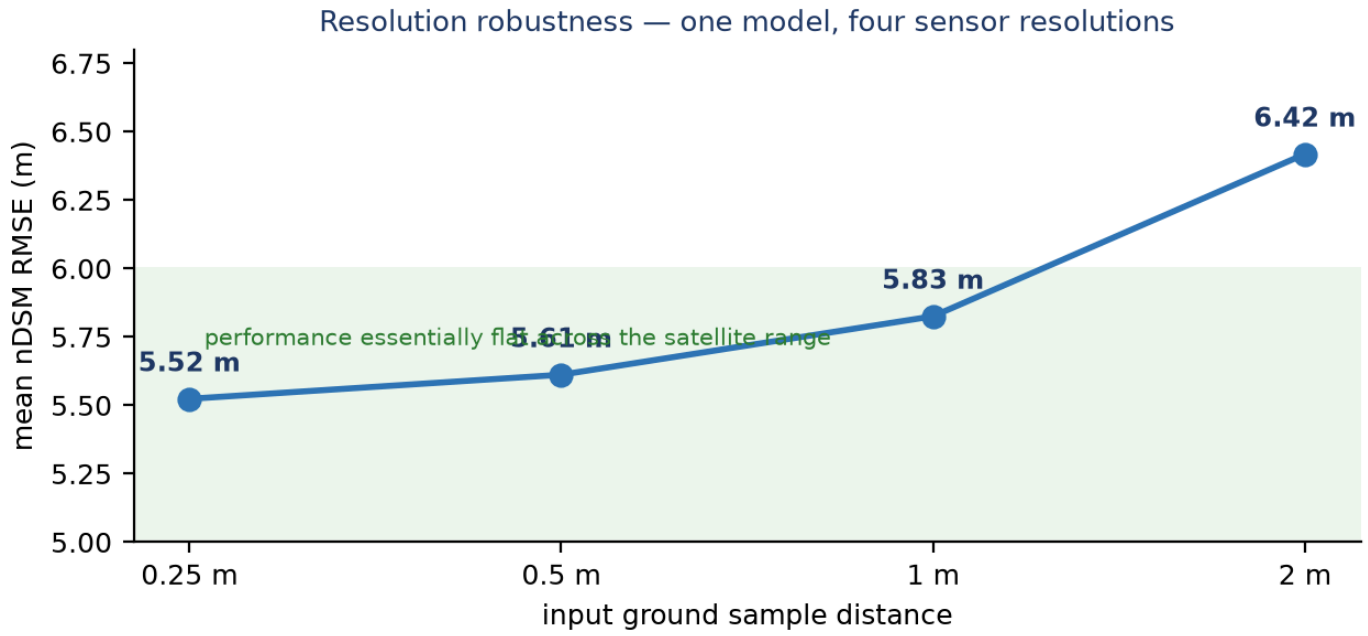



*Figure 5 — Accuracy against input resolution. Flat from 0.25 m to 1 m; graceful, not catastrophic, at 2 m.*

| Input GSD | Mean RMSE | Change vs 0.25 m | Comparable sensor |
|---|---|---|---|
| 0.25 m | **5.52 m** | +0.00 m | aerial / drone orthophoto |
| 0.5 m | **5.61 m** | +0.09 m | WorldView-class panchromatic |
| 1 m | **5.83 m** | +0.30 m | Cartosat-2 / high-res commercial |
| 2 m | **6.42 m** | +0.89 m | Cartosat-1 / Sentinel-2 upsampled |

Achieved by normalising every input to the 0.33 m training GSD before inference, so the model always sees features at the scale it was trained on. Upsampling is capped at 3× so we never invent detail that is not there.

Generated 29 September 2026 from the shipped checkpoint `decoder_best.pth` (Depth Anything V2 Base + multi-scale DPT head, epoch 28). Every accuracy figure in this document is read from `outputs/benchmark_ahn/results.json` in the repository and can be regenerated with `python scripts/benchmark_ahn.py`. Resolution figures from a dedicated sweep of the same model. Repository: github.com/Aryan241/Heimdall